

Management external payload

Scheme	External payload B
Our proposed DPKI	1615
Centralized PKI	1262
Multi-CA based DPKI	1674
full-contract DPKI	644

Intra-domain authentication external payload

Scheme	External payload B
Our proposed DPKI (off-chain)	1585
Our proposed DPKI (on-chain)	2342
Centralized PKI	384
Multi-CA based DPKI	3286
full-contract DPKI	925

Cross-domain authentication external payload

Scheme	External payload B
Our proposed DPKI	3925
Centralized PKI	1570
Multi-CA based DPKI	3740
full-contract DPKI	1298

On-chain primitive cost summary

Primitive	Used by	On-chain storage B	Gas (10^3)
Cert/root write	Proposed Mgmt.; Multi-CA Mgmt.; full-contract Mgmt.	654	235.7
Auth record write + read	Proposed Intra-on; Proposed Cross	590	281.5
Full-contract intra auth	full-contract Intra-on	942	357.2
Full-contract cross auth	full-contract Cross	1006	363.3